A Short Guide to the Fourier Transform

1. Why frequency matters

Many signals are easier to understand in terms of the frequencies they contain than as values changing over time. The Fourier transform moves a signal from the time domain to the frequency domain, where filtering, modulation and noise become simple to describe.

2. Definition

For a signal g(t), the Fourier transform is defined as

$$G(f) = \int g(t)\, e^{-j2\pi ft}\, dt$$

The magnitude |G(f)| tells us how strongly each frequency is present, while the phase of G(f) tells us how that component is shifted in time.

3. Useful properties

1. Linearity: the transform of a sum is the sum of the transforms.

2. Time shift: delaying a signal by t_0 multiplies its transform by $e^{-j2\pi ft0}$; the magnitude does not change.

3. Scaling: compressing a signal in time spreads its spectrum in frequency.

4. Common pulses

Pulse	Time domain	Spectrum
Rectangular	rect(t/T)	T sinc(fT)
Triangular	tri(t/T)	T sinc²(fT)
Gaussian	exp(–πt²)	exp(–πf²)

A narrow pulse has a wide spectrum and a wide pulse has a narrow one. This trade-off appears again whenever we design filters or choose a sampling rate.

5. Sampling and aliasing

A computer only sees a signal at discrete instants, spaced $T_s = 1/f_s$ apart. Sampling makes copies of the spectrum repeat every f_s along the frequency axis. If the signal contains no frequencies above B, the copies do not overlap as long as

$$f_s \geq 2B$$

This is the Nyquist criterion. When it is violated, high frequencies fold back and appear as lower ones, an effect called aliasing. Once aliasing has happened it cannot be undone, so real systems place a low-pass filter before the sampler.

6. A worked example

Speech carries most of its information below about 3.4 kHz. Telephone systems therefore filter speech to that band and sample it at 8 kHz, which leaves a small guard band above the Nyquist rate of 6.8 kHz. Music needs the full audible range up to 20 kHz, which is why CD audio is sampled at 44.1 kHz.

Application	Bandwidth B	Sampling rate fs
Telephone speech	3.4 kHz	8 kHz
CD audio	20 kHz	44.1 kHz
Wideband audio	7 kHz	16 kHz

7. Summary

1. The Fourier transform describes a signal by the frequencies it contains.
2. Shifting a signal in time changes only the phase of its spectrum.
3. Narrow in time means wide in frequency, and the other way round.
4. Sample at least twice the highest frequency, and filter first to prevent aliasing.